

1 Background

Best Match Graphs and Gene Families. Important aspects of the evolutionary history of a gene family is described by a leaf-colored tree as follows: Let T be a rooted phylogenetic tree (i.e., a tree without internal vertices with a single descendant) with leaf-set L denoting the genes. Moreover, we assume that for each gene $u \in L$ we know the species $\sigma(u)$ in which it appears. Given (T, σ) we can now ask for the closest relatives of a gene $x \in L$ in all other species $\sigma' \neq \sigma(x)$. To this end, we need the notion of the last common ancestor of two genes, $\text{lca}(x, y)$, which is defined as the (unique) vertex in T that is an ancestor of both x and y and has no descendant with the same property. Let us write $\preceq$ for the partial order on $V(T)$ defined by $x \preceq y$ if y lies on the unique path from the root to x . W.r.t. $\preceq$ there the leaves are the minimal elements, the root is the unique maximal element and $\text{lca}(x, y)$ is the $\preceq$ -minimal vertex in $V(T)$ satisfying both $x \preceq \text{lca}(x, y)$ and $y \preceq \text{lca}(x, y)$.

For a gene $x \in L$, a *best match* is a gene $y \in L$ with $\sigma(x) \neq \sigma(y)$ such that for all genes $y' \in L$ with $\sigma(y) = \sigma(y')$ we have $\text{lca}(x, y) \preceq \text{lca}(x, y')$. The notion of a best match makes the idea of a most closely related genes in different species mathematically precise. This *best match relation* can be represented as a vertex colored graph $G(T, \sigma)$ with vertex set $V(G) = L$, vertex colors $\sigma(x)$ for all $x \in L$, and directed edges $(x, y) \in E(G)$ if and only if y is a best match for x . Note that by definition every gene x with color $\sigma(x)$ has a best match in every other species, i.e., for all colors/species σ' , there is a best match y of x with $\sigma(y) = \sigma'$. A graph with this property is called *color sink-free*.

A vertex-colored graph (G, σ) is called a *best match graph* (BMG) if there is a leaf-colored tree (T, σ) such that (G, σ) corresponds to the best match relation derived from (T, σ) . Clearly, any BMG is color-sink-free. Not all color-sink-free graphs (G, σ) derive from a tree, however.

From (G, σ) one can infer a sets of informative and forbidden triples,

$$\begin{aligned} R &:= \{ab|b' : \sigma(a) \neq \sigma(b) = \sigma(b'), (a, b) \in E(G), (a, b') \notin E(G)\} \\ F &:= \{ab|b' : \sigma(a) \neq \sigma(b) = \sigma(b'), b \neq b', (a, b), (a, b') \in E(G)\} \end{aligned}$$

Triples $ab|c$ correspond to trees of the form $((ab)c)$. Efficient algorithms available to check if a pair (R, F) of sets of is consistent in the sense of deriving from a common tree so that T contains all triples in R and none of the triples in F .

A graph (G, σ) is a BMG if and only if (G, σ) is color-sink-free and (R, F) is consistent. In this case, the so-called Aho-tree T_R of R is the unique least resolved tree explaining (G, σ) .

Phylogenetic Networks. In recent years there has been increasing interest in phylogenetic networks (directed acyclic graphs with a single “root” and no interior vertices with a single descendant) as model of phylogeny. Most commonly, networks instead of trees are used in order to include hybridization or as means of representing uncertainty. As in the case of trees, there is a partial order $\preceq$ encoding reachability from the root: $x \preceq y$ iff y lies along a (no longer

unique) path from the root to x . Again generalizing the situation in trees we call a $\preceq$ -minimal vertex in a network N a *leaf* and write L for the set of leaves. For a subset $A \subset L$ we can define its least common ancestors (plural, since there is in general more than one) as the set $\text{LCA}(A)$ consisting of the $\preceq$ -minimal vertices z satisfying $x \preceq z$ for all $x \in A$. Since the root is an ancestor of all leaves, $\text{LCA}(L) \neq \emptyset$.

Now we are in the position to define best matches for leaf-labeled phylogenetic networks (N, σ) : we say y is a best match of x if $\sigma(y) \neq \sigma(x)$ and for all $y' \in L$ with $\sigma(y') = \sigma(y)$ there is $u \in \text{LCA}(x, y')$ such that $u \prec v$ for some $v \in \text{LCA}(x, y)$. Equivalently, for all y' with $\sigma(y') = \sigma(y)$ and all $u \in \text{LCA}(x, y')$ and $v \in \text{LCA}(x, y)$ holds $u \not\prec v$.

In analogy to the case of trees, we may ask when a graph (G, σ) can be obtained as the best match graph of a network (N, σ) .

Best match graph for networks were studied recently in still unpublished work by Patricia A. Ebert and Marc Hellmuth (U. Stockholm).

Not surprisingly, many more colored graph (G, σ) can be explained by networks (N, σ) than by trees (T, σ) . We say that (G, σ) is a network-BMG if there is a phylogenetic network (N, σ) such that (G, σ) represents the best matches in (G, σ) .

Let (G, σ) be a colored digraph. We call a vertex $v \in V$ a *single color representative (sikor)* if v is the only vertex with color $\sigma(v)$. A vertex $v \in V$ is an *in-hub* if all vertices u with a different color $\sigma(u) \neq \sigma(v)$ are connected to v by an edge $(u, v) \in E$. If each sikor vertex in (G, σ) is an in-hub, then (G, σ) has the *sikor-in-hub property*. It is not difficult to check that color-sink-free implies sikor-in-hub.

A graph (G, σ) is a network-BMG if and only if (G, σ) is sikor-in-hub.

The proof is constructive. In the positive case the network (N, σ) is readily computed. To this end we first generalize the notion of a cherry: a pair of leaves $\{x, y\}$ is a bicolored cherry if (i) x and y have a common parent and (ii) $\sigma(x) \neq \sigma(y)$.

The first step is to construct the BIC-cherry-network for L . It consists of the root ρ , the leaves L , and a set of intermediary vertices p_{xy} for $x, y \in L$ with $\sigma(x) \neq \sigma(y)$. This vertex has x and y as its children and ρ as its unique ancestor. (If $|L| = 2$, p_{xy} takes on the role of the root.)

The second step consists in an expansion of the network. To this end we define the *extension* $[xy : xz]$ a cherry (x, y) with parent p_{xy} and a third vertex $z \in L$ as the insertion of a new vertex q_{xz} and the subsequent insertion of the directed edges (p_{xy}, q_{xz}) , (q_{xz}, x) and (q_{xz}, z) .

Given a graph (G, σ) with the sikor-in-hub property, proceed as follows:

- (1) Initialize (N, σ) as the BIC-cherry network on L with root ρ .
- (2) For all $x, y \in L$ with $(x, y) \notin E(G)$ and $\sigma(x) \neq \sigma(y)$ extended (N, σ) by $[xy : xy']$ for an arbitrarily chosen $y' \in L \setminus \{y\}$ with $\sigma(y) = \sigma(y')$.

Ebert and Hellmuth show that the final network (N, σ) resulting from this procedure explains the input graph.

From an application point of view, however, the problem is that (N, σ) is almost never a tree, even if (G, σ) is in fact a tree-BMG. This begs the question whether we can stepwisely modify (N, σ) in such a way that it still explains (G, σ) but becomes more “tree-like”.

Two definitions of best matches for networks

Recall the “standard” strict definition of best matches in a leaf-colored network (N, σ) with leaf set $X = L(N)$.

Definition. $y \in X$ is a *best match* of a $x \in X$ in (N, σ) if $\sigma(x) \neq \sigma(y)$ and $u \not\prec_N v$ for all $y' \in X$ with $\sigma(y') = \sigma(y)$ and all $u \in \text{LCA}(x, y')$ and all $v \in \text{LCA}(x, y)$.

There is a useful alternative, weaker definition of best matches. For all $x \in X$ and $\tau \in \sigma(X)$, $\tau \neq \sigma(x)$ define

$$Q(x, \tau) := \min_{\preceq} \{v \in \text{LCA}(x, y) : \sigma(y) = \tau\} \quad (1)$$

That is, $Q(x, \tau)$ is the set of all lowest common ancestors between x and a leaf with color τ .

Definition. $y \in X$ is a *weak best match* of $x \in X$ in (X, σ) if $\sigma(x) \neq \sigma(y)$ and $\text{LCA}(x, y) \cap Q(x, \sigma(y)) \neq \emptyset$.

This is, $y \in X$ is a weak best match if it is reachable from x via one of the the lowest common ancestors in x .

For trees, best matches and weak best matches are the same and coincide with the definition of best matches on trees.

2 Work Program

2.1 Objectives

- (1) Compare the definition of best matches and weak best matches:
 - (a) Check whether every best match in (N, σ) is also a *weak best match*. Test this computationally, but also try to give a formal argument.
 - (b) Check if a modified version of the BIC-cherry + expansion procedure yields networks that explain weak best match graphs.
- (2) Find a way to edit the explaining networks for (weak) best matches to become more tree like.

2.2 Tasks

- (0) Use **AsymmeTree** to generate trees. Modify the resulting trees by inserting hybridization vertices to generate phylogenetic networks (iteratively choosing two random points on edges of the tree/network and connecting them).
- (1) Construction of explaining networks:
 - (a) Implement the BIC-cherry+expansion algorithm for given graphs (G, σ) as in the tech report from Patricia Ebert and Marc Hellmuth.
 - (b) Modify this algorithm by restricting expansions $[xy : xz]$ to accepting edges xz in graph (G, σ) and possibly relabel q_{xz} as p_{xz} .
 - (c) Thoroughly test these algorithms! Think of unit tests for your implementations!
 - (d) Test if variant (b) always yields an explanation for weak best matches: to this end, generate a network (N, σ) , compute its *weak best match graph*, run variant (b) to produce an explaining network (N', σ) , and check that the weak best match graph of (N', σ) coincides with the weak best match graph of (N, σ) . If this turns out to be **not** true, find the minimal examples where the networks differ.
- (2) Simplification of explaining networks
 - (a) Construct tree-BMGs (G, σ) from (**AsymmeTree**-generated) trees and compute their least resolved trees T^* . This is the target.
 - (b) Take the tree-BMGs from (a) and construct the BIC-cherry+expansion explanations (N, σ) .
 - (c) Implement graph editing by “pulling up” or “pulling down” edges, and removing vertices with the same parents and children, as described in class.
 - (d) Check both for single moves and combinations of a small number of moves whether the modified network (N', σ) has the same (weak) best match graph as (N, σ) .
 - (e) Devise heuristics to search for edit paths that allow the stepwise conversion of the BIC-cherry+expansion explanations of (N, σ) in the least resolved tree-explanation (lrt) T^* .
 - (f) If this is not always successful, find minimal examples that fail, try to find out why they fail, and if possible use this information to improve the heuristic for editing schedules.
- (3) Interpretation: Try to find out why and where simple approaches fail (as e.g. in 2f) and make recommendation for further improvements/modifications.

3 Exam

Groups will be examined together. The exam will consist of individual questions for each group member, followed by a joined presentation. Please summarize your setup with a focus on how the correctness of modules was verified. This is not meant to be a code review! Summarize all scientific finding appropriately, following the guidelines of the work program. It is highly encouraged that you prepare slides for this purpose.